University of Pisa
Department of Data Science and Business Informatics

The Rise of Oceanus Folk

Interactive Visual Analytics of a Music Knowledge Graph

Course	Visual Analytics 602AA
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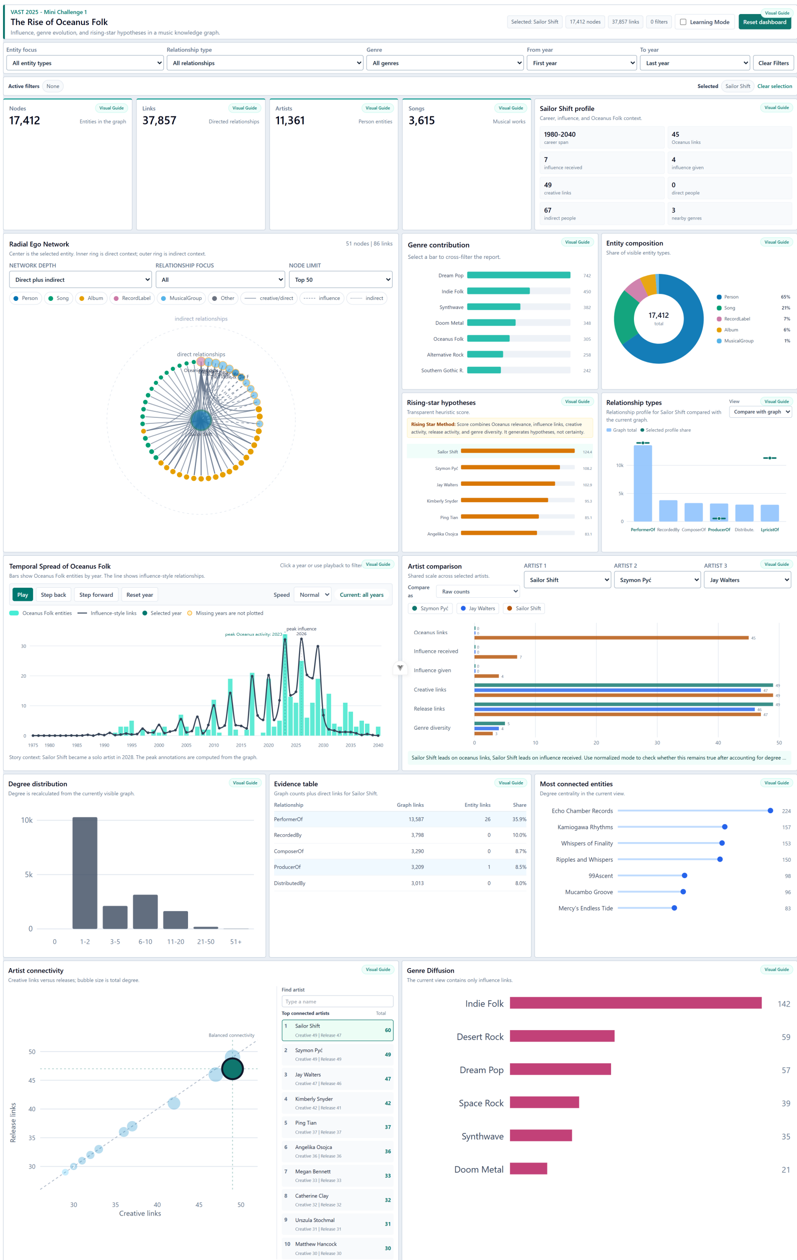
Project	VAST Challenge 2025, Mini Challenge 1
Repository	github.com/MuawiyaNasser/oceanus-folk-visual-analytics

Project summary. The system helps a journalist investigate how Sailor Shift, artists, works, genres, and typed music relationships connect to the rise of Oceanus Folk. Computational summaries prepare the evidence, while linked filtering and selection let the analyst form and verify hypotheses.



Academic Year 2025/2026

1 Introduction and Data

The project studies the music knowledge graph released for VAST Challenge 2025 Mini Challenge 1, which asks analysts to investigate Sailor Shift and the rise of Oceanus Folk. The graph contains **17,412 entities** and **37,857 directed relationships**. Entities include people, songs, albums, record labels, musical groups, and genre-related nodes. Typed links describe performance, recording, composition, production, distribution, membership, covers, samples, references, and stylistic connections.

A table alone cannot communicate this relational structure, while drawing the complete graph would create an unreadable node–link hairball. The dashboard therefore combines a focused ego network with coordinated statistical views to communicate influence, collaboration, genre diffusion, temporal evolution, artist comparison, and rising-star hypotheses. The dataset does not provide reliable geographic fields, so "spread" is interpreted as diffusion through the music ecosystem rather than geographic movement.

Data element	Examples	Dashboard use
Entities	People, songs, albums, labels, groups	Network nodes, filters, composition, rankings
Relationships	PerformerOf, RecordedBy, ComposerOf, ProducerOf	Network edges, evidence, relationship comparison
Attributes	Name, entity type, genre, year	Labels, grouping, colour, temporal filtering
Derived measures	Degree, creative links, releases, genre diversity	Comparison, distribution, hypotheses

2 Analytical Goals and Design

The principal goal is to understand how Oceanus Folk develops through artists, works, genres, and influence. The interface supports the official MC1 tasks: profiling Sailor Shift, showing influence over time, comparing direct and indirect collaborations, identifying Oceanus Folk impact, distinguishing gradual from intermittent rise, finding affected artists and external genres, comparing three artists, and producing transparent hypotheses for future stars.

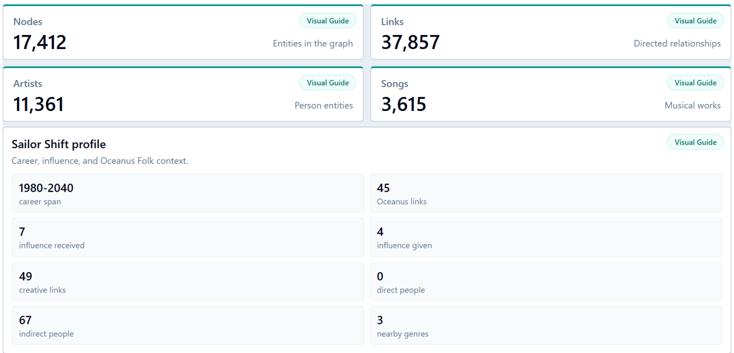
The design follows *overview first, filter and select, details on demand.* KPI cards and the Sailor Shift profile establish context. The radial network receives the largest area because relationships are central to the task. Position and length support quantitative comparison; colour hue separates nominal entity types; line and column marks encode time; a donut is reserved for small part-to-whole composition; and tables are used only for exact verification. Teal is the selection accent; blue supports factual comparison; orange signals provisional hypotheses; slate represents neutral structure; rose identifies influence evidence; and the entity palette remains consistent across views. Learning Mode and per-card Visual Guides explain purpose, meaning, interaction, and reading without forcing long text into the default dashboard.

Visual Analytics model. The application calculates graph metrics, filtered subsets, rankings, and temporal summaries. The analyst then uses linked visual evidence to compare alternatives, challenge a hypothesis, and return to the complete view through reversible interactions.

3 Data Preparation

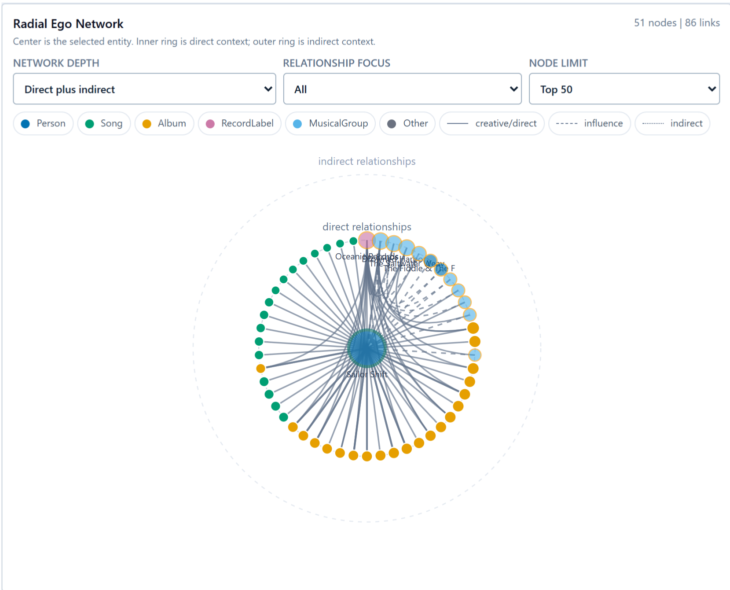
The frontend loads the JSON graph, normalizes identifiers and missing values, extracts years, indexes nodes and links, and calculates degree, in-degree, out-degree, neighbours, relationship counts, timeline aggregates, genre aggregates, and artist-level measures. Unknown metadata remains explicit instead of being silently invented. Entity Focus and Genre Focus are applied after graph context is computed, so bridge nodes are preserved instead of breaking paths. The ego network applies a node limit and prioritizes nearby, highly connected entities so that the local graph remains readable.

4 Dashboard Visuals



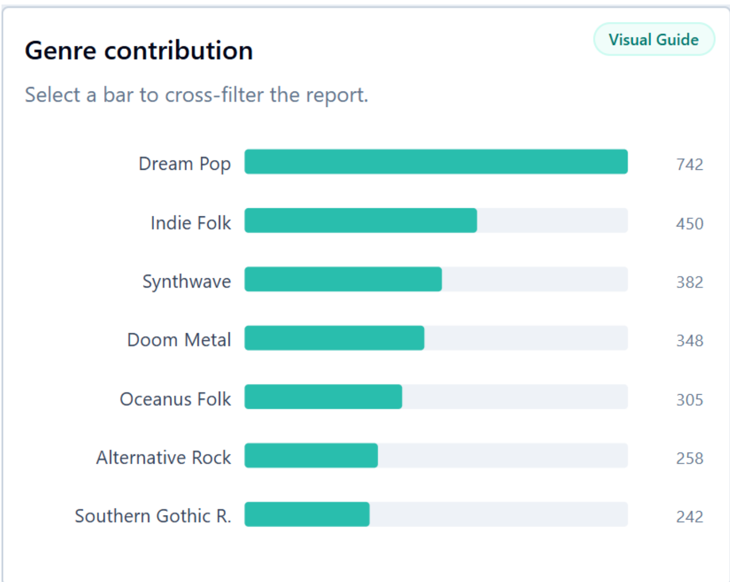
KPI cards and Sailor Shift profile

Cards are used because exact headline values must be scanned faster than they could be read from a chart. The KPI cards summarize visible nodes, links, artists, and songs and recalculate after filtering, implementing *overview first*. The Sailor Shift profile adds career span, Oceanus links, influence received and given, creative links, direct and indirect people, and nearby genre diversity, directly supporting the MC1 career-profile task. White surfaces and dark text maximize legibility, while the thin teal rule links the overview to the dashboard's selection accent.



Radial ego network

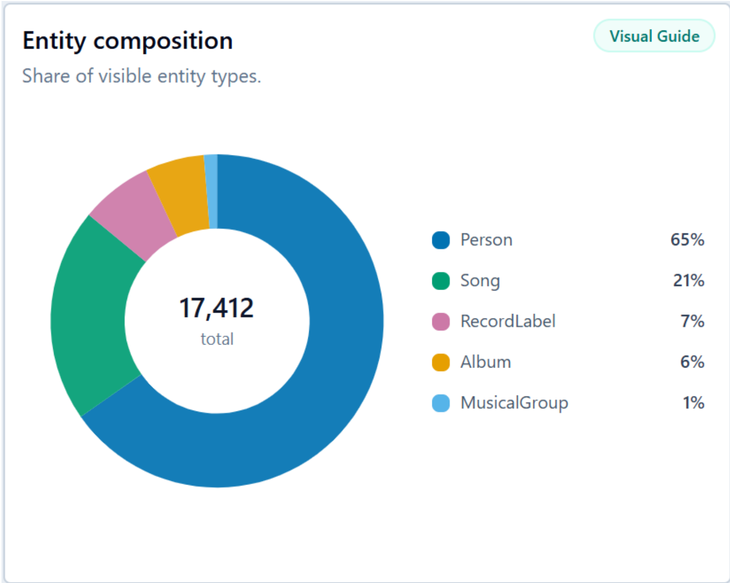
A radial ego network is used because the task concerns relationships around one focus entity, and rings communicate direct versus indirect distance more clearly than a free force layout. The centre is selected; **Network Depth**, **Relationship Focus**, and **Node Limit** control scope and clutter. A consistent colour-blind-aware categorical palette maps people to blue, songs to green, albums to amber, labels to pink, groups to sky blue, and other types to gray. Teal outlines emphasize selection, amber outlines flag unknown genre metadata, node size reflects degree, and restrained gray edge styles distinguish direct, influence, and indirect relations. Hover and click provide details and refocusing.



Genre contribution

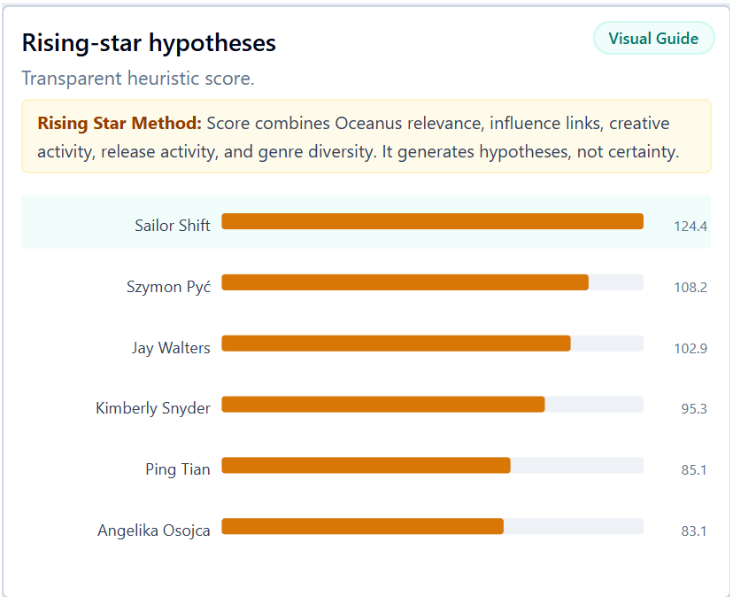
A sorted horizontal bar chart is used because this is a categorical ranking with long genre names; aligned length on a common baseline supports accurate comparison. Teal bars connect the view to genre selection and Oceanus-related exploration, while pale gray tracks preserve the full comparison range and make filtering changes visible. The darker teal active bar is consistent with the dashboard selection colour. Clicking a genre cross-filters linked views, clicking it again clears it, and the ranking updates after other filters.

5 Dashboard Visuals Continued



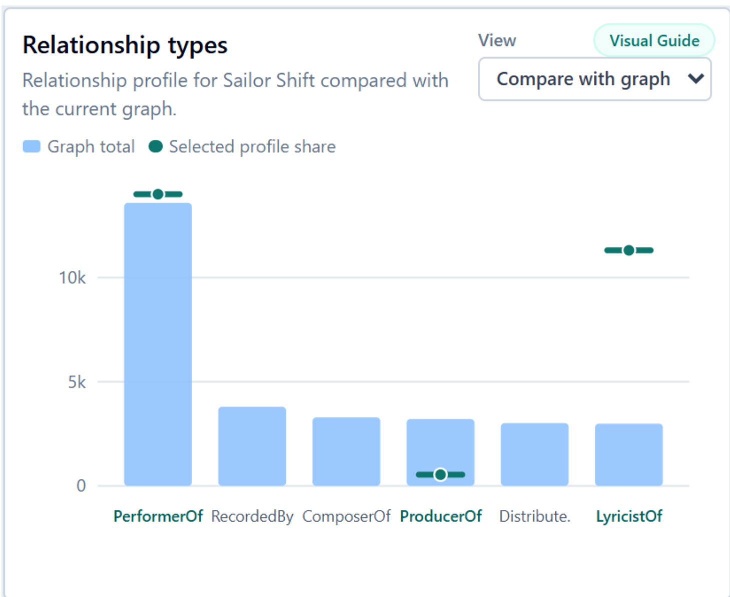
Entity composition

A donut is appropriate because a small number of entity categories partition one meaningful whole. The centre reports the visible total, while slice angles and percentages communicate composition. Colours deliberately match the network: people are blue, songs green, labels pink, albums amber, and groups sky blue. This cross-view consistency reduces relearning and uses hue only for nominal categories. Filtering updates the centre value, slices, and legend to show which entity type dominates the current context.



Rising-star hypotheses

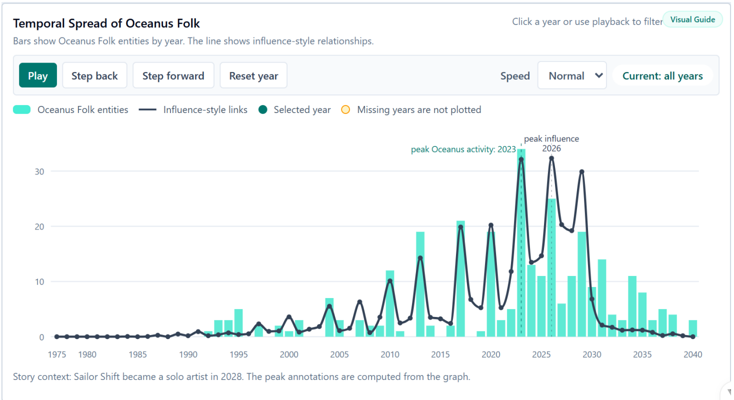
A sorted horizontal bar chart is chosen because candidates must be ranked on one transparent heuristic score, and aligned lengths are easier to compare than areas or angles. The score is calculated as: creative links $\times 2$, influence links $\times 1.5$, nearby genre diversity $\times 1.25$, recent activity $\times 3$, Oceanus relevance $\times 3$, closeness to Sailor Shift, $\log(1 + \text{degree})$, and a small penalty for already-notable artists. Orange is used deliberately as a cautionary, provisional colour: the score generates hypotheses and is not established truth. Selecting a candidate updates the network and artist comparison so the ranking starts an investigation rather than ending it.



Relationship types

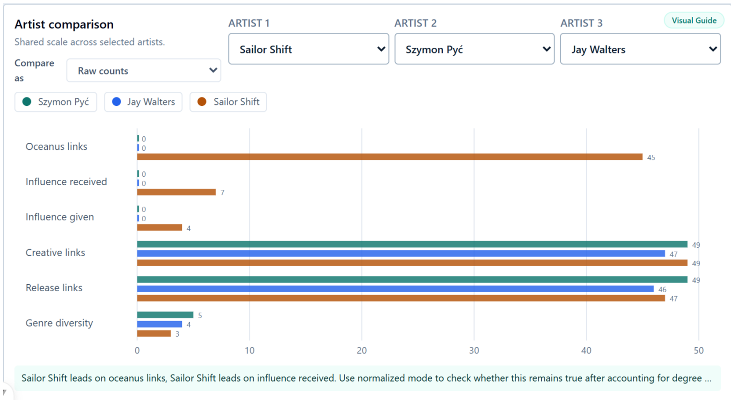
A column chart is suitable because the relationship names are short and the goal is comparison of discrete counts from a shared zero baseline. Column height encodes the frequency of PerformerOf, RecordedBy, ComposerOf, ProducerOf, DistributedBy, and LyricistOf. Calm blue represents factual structural evidence rather than selection or warning, while teal is reserved for an active category. The chart updates after filtering and reveals which link type dominates the current context.

6 Dashboard Visuals Continued



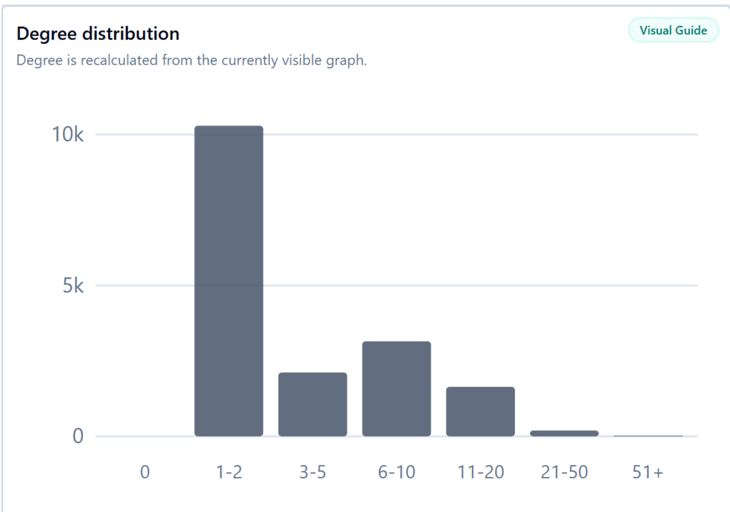
Temporal Spread of Oceanus Folk

A combined column-and-line chart is used because it compares two temporal signals without hiding their different mark types: aqua columns show yearly Oceanus Folk activity and a dark navy line shows influence-style relationships. Aqua connects activity to the dashboard's Oceanus accent; navy provides strong contrast for the second measure; dark teal identifies the selected year; and amber marks missing-year uncertainty. Time runs left to right and peak annotations guide interpretation. Click, playback, step controls, speed, and reset coordinate the selected year with the other views.



Artist comparison

Grouped horizontal bars are used because three artists must be compared repeatedly across the same measures on shared axes. Teal, blue, and burnt orange give each artist a stable, distinguishable identity across every metric without implying good or bad performance; neutral gridlines keep attention on bar length. Measures include Oceanus links, influence received and given, creative links, release links, and genre diversity. Raw counts show absolute evidence, while normalization by degree or releases supports fairer comparison. Selectors replace artists and chart selection can change the active focus.



Degree distribution

A histogram is the appropriate choice because degree is quantitative and the question concerns its distribution, not a ranking of named entities. Bins reduce thousands of individual degrees into readable intervals and reveal many low-degree entities versus a few hubs. Slate is intentionally neutral because this chart describes background graph structure rather than an active category; its subdued tone prevents the large first bin from dominating the dashboard. The histogram recalculates after filtering.

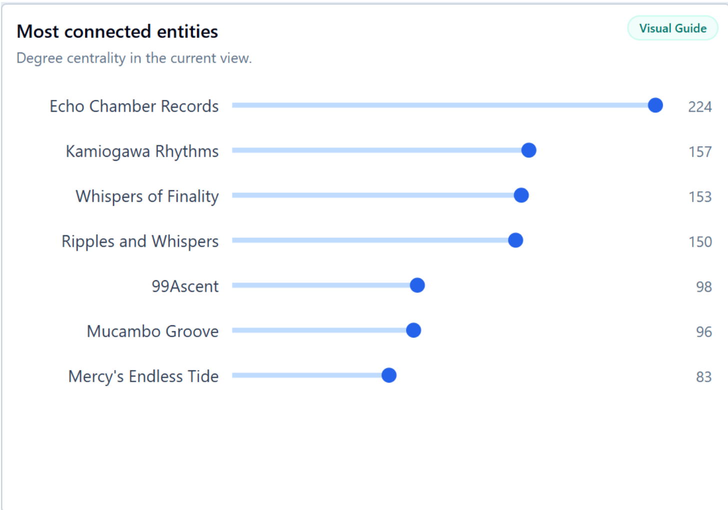
7 Dashboard Visuals Continued


Evidence table
Visual Guide
Graph counts plus direct links for Sailor Shift.

Relationship	Graph links	Entity links	Share
PerformerOf	13,587	26	35.9%
RecordedBy	3,798	0	10.0%
ComposerOf	3,290	0	8.7%
ProducerOf	3,209	1	8.5%
DistributedBy	3,013	0	8.0%

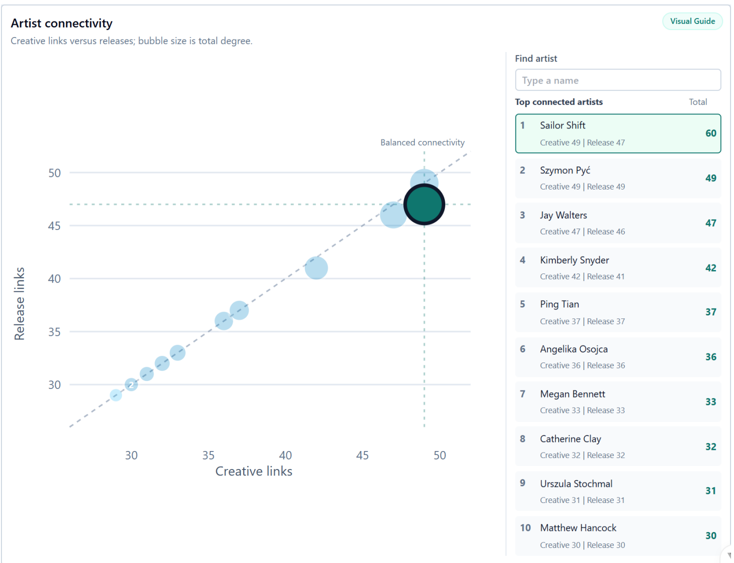
Evidence table

A compact table is used instead of a chart because the task here is verification of exact relationship counts and shares. It complements the pattern-oriented visuals and remains intentionally short. Colour is minimized: dark text carries the values, light gray rules organize rows, and the white surface avoids assigning unsupported semantic meaning to categories. This restrained treatment keeps exact evidence readable without competing with interactive selections.



Most connected entities

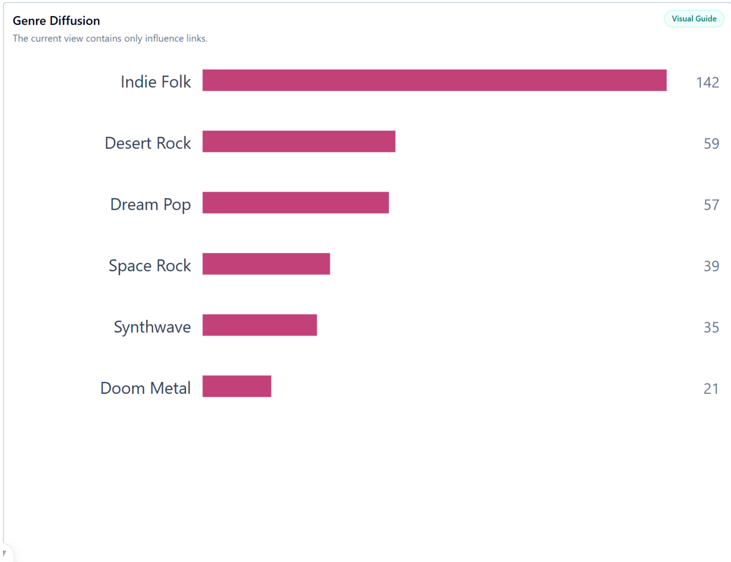
A lollipop chart is chosen for a compact degree ranking: aligned dot positions preserve bar-chart accuracy while using less ink. Entities are sorted so the strongest hubs appear first. Blue dots communicate factual centrality consistently with other structural comparisons, pale blue stems provide scale context, and teal marks the selected entity. Clicking a dot moves from the ranking to that entity's local network evidence.



Artist connectivity

A bubble scatterplot is used because two quantitative variables must be compared simultaneously: creative links are mapped to horizontal position and release links to vertical position. The axes focus on the actual visible range instead of starting at zero, which is acceptable here because position, not bar length, carries the quantitative meaning. Bubble size gives a subtle third encoding for total degree, while exact shared coordinates are grouped with a small count badge rather than displaced with misleading jitter. Sky blue is used for comparable artist points, teal marks the selected artist, white outlines separate overlaps, and the dashed diagonal shows balanced creative and release activity. The ranked Top 10 panel on the right keeps full artist names readable, supports search, and links back to the scatterplot through hover and click selection.

8 Dashboard Visuals Continued



Genre Diffusion

A horizontal bar chart ranks external genres connected to Oceanus Folk, which gives long labels room and uses aligned length for accurate comparison. The aggregation assigns every link to exactly one class: Influence, Creative, or Other. The current data contains only Influence links for the displayed genres, so zero-value categories are hidden and the chart states this limitation. Rose is reserved for influence evidence, distinguishing it from teal selection and orange creative evidence. Clicking a genre applies the linked genre filter.

9 Interaction Model

All views share one understandable analytical state: selected entity, entity focus, relationship type, genre, year range, and degree range. Network depth and node limit control only the local graph presentation. A chart selection is highlighted, related views update, a repeated click clears the selection where applicable, and **Reset dashboard** restores Sailor Shift and the full dataset. Learning Mode opens all Visual Guides for presentation; each Visual Guide can also be opened or closed independently for details on demand.

Action	Coordinated response	Analytical purpose
Click a node or ranked entity	Network recentres; details and comparison update	Move from overview to local evidence
Click a genre or relationship	Linked charts and graph subset update	Isolate one categorical explanation
Click or play a year	Applicable views recalculate for that period	Examine temporal development
Change artist selectors or mode	Shared measures are redrawn	Compare absolute and normalized profiles
Reset filters	Full graph state and Sailor Shift return	Keep exploration reversible
Visual Guide or Learning Mode	Explanation panels open locally or globally	Support first-time usability and oral presentation

10 State of the Art and Original Contribution

The dashboard follows coordinated-view practices common to Tableau, Power BI, and other visual analytics systems: compact controls, KPI feedback, linked charts, highlighting, and details on demand. Graph systems such as Gephi, Cytoscape, and Neo4j Bloom support network exploration, while Observable/D3 examples often demonstrate custom graph layouts. This project differs by combining graph exploration, temporal analysis, genre diffusion, artist comparison, centrality evidence, and rising-star hypotheses in one task-oriented interface. The original D3 contribution is the radial ego network, which replaces an unconstrained force layout with explicit distance rings, relationship controls, degree-based sizing, entity focus, genre context, and a readable subset rule that avoids the large-graph hairball problem.

11 Analytical Use Case

A music journalist begins with the KPI overview and the Sailor Shift Profile to understand the visible graph and Sailor's role. The analyst changes Relationship Focus to influence, examines direct and indirect neighbours in the radial network, and selects a peak period in the timeline to test whether the rise is gradual or intermittent. Genre Contribution and Genre Diffusion reveal external stylistic connections. The analyst then selects the top three Rising-Star Hypotheses, compares them with Sailor Shift in raw and normalized modes, and checks centrality, artist connectivity, and exact evidence counts. The conclusion is not a deterministic prediction, but a defensible short list of likely future Oceanus Folk stars supported by coordinated visual evidence.

12 Implementation

The project uses Vue, JavaScript, CSS, SVG, and D3. The code flow remains deliberately simple: load and normalize the graph, hold the shared dashboard state, apply filters, compute derived metrics, and render reusable visual components. D3 scales map values to position, length, radius, and colour. Vue propagates state changes so that one interaction updates all relevant components.

13 Limitations

Some entities lack reliable genre or year metadata. Unknown values remain visible or are excluded only where explicitly stated, so missing data can affect rankings and temporal interpretation. Degree measures connectivity, not artistic quality. Influence-style links reflect the available graph relationships and do not prove causality. The rising-star score is therefore an explainable hypothesis generator, not a forecast with statistical certainty. The dashboard uses focused subsets to reduce cognitive overload, which means that the radial network is intentionally not a complete graph view.

14 Conclusion

The project turns a large music knowledge graph into a compact analytical workspace. Computational summaries expose candidate patterns, while linked interaction lets the user test them across relationships, time, genres, and artist profiles. Appropriate visual variables, consistent scales, reversible filtering, and details on demand reduce cognitive load and support human-in-the-loop reasoning.

Acknowledgement of AI-Assisted Development

OpenAI Codex was used as a development assistant for implementation, debugging, refactoring, visual quality checks, and report preparation. The analytical direction, project decisions, interpretation, and final responsibility remain with the author.

Final point. This is a Visual Analytics system rather than a static collection of charts: computation prepares evidence, coordinated visualization exposes patterns, and interaction lets the analyst refine and verify an interpretation.